MIL-A-28768/2A(EC)
15 December 1972

SUPERSEDING
MIL-A-28768/2(EC)
15 April 1971
(See 6.5)

MILITARY SPECIFICATION

ANTENNA, FIXED HIGH FREQUENCY, HIGH TAKE-OFF ANGLE

1. SCOPE

1.1 Scope. This specification covers specific requirements for an antenna capable of producing high angle radiation (hereinafter referred to as the antenna), to be used at shore stations for short range skywave communications in the high frequency region of the radio spectrum.

1.2 Classification. The antenna shall be of the following types, classes, and polarizations, as specified (see 6.2).

Type	Covers the frequency range of:
I	2 MHz - 8 MHz
II	2 MHz - 20 MHz
III	2 MHz - 30 MHz
IV	3 MHz - 12 MHz
V	3 MHz - 20 MHz
VI	3 MHz - 30 MHz
VII	4 MHz - 15 MHz
VIII	4 MHz - 30 MHz

Class	Wind and Ice conditions
1	100 mph (no ice)
2	125 mph (no ice) or 90 mph with 1/2 inch radial ice on all members
3	160 mph (no ice) or 90 mph with 1/2 inch radial ice on all members
4	Special case, as defined in the procurement document

Polarization (see 6.3)	Orientation of E vector
RHC	Right hand circular
LHC	Left hand circular
H	Linear

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2. APPLICABLE DOCUMENTS

2.1 The following document of the issue in effect on date of invitation or bids or request for proposal, forms a part of the specification to the extent pecified herein.

SPECIFICATION

MILITARY

MIL-A-28768 - Antennas, Fixed High Frequency General Specification For

(Copies of specifications, standards, drawings, and publications required y suppliers in connection with specific procurement functions should be obtained rom the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. The antenna shall comply with MIL-A-28768 and the requirements specified herein. Requirements shall be met individually, simultaneously, nd under all possible combinations. The antenna shall be furnished as a complete system, ready for installation and operation. It shall be designed for roadband operation without tuning or adjustment.

3.2 Classification of requirements. Requirements for the antenna are lassified herein as follows:

Requirement	Paragraph
Size	3.3
Environmental conditions	3.4
Structural	3.5
Electrical performance	3.6
Ground System	3.7

3.3 Size. The antenna shall comply with the size limitations listed elow. All dimensions are in feet.

Maximum length, width or diameter, including guys	400
Maximum height	150

3.4 Environmental conditions. The antenna shall meet the environmental equirements specified in MIL-A-28768. The antenna shall be designed to meet he allowable stress requirements specified in MIL-A-28768 when loaded with the ind and ice conditions defined by the Class specified (see 1.2). Unless otherise specified in the procurement document, Class 2 shall apply (see 6.2). The ord "or" in the Class specified means that the antenna must meet both conditions ndependently. Structural relief devices (fuses) shall not be used.

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3.5 Structural requirements.

3.5.1 Rotational stability. The antenna shall be designed with an appropriate guying or bracing system which will eliminate the possibility of any rotational instability. The antenna shall be designed such that effects of any force resulting from the required wind and ice loading will not cause a rotation about a vertical axis through the center of the antenna greater than 30 minutes of arc.

3.5.2 Field adjustment of cable tension. The design of the antenna should provide facilities for field adjustment of cable tensions both during and after erection. If a peripherial cable is used, each span of the peripherial cable between adjacent poles or other points of attachment shall have a turnbuckle or similar cable adjustment device.

3.5.2.1 A detailed description of the proper method of initial field installation and subsequent maintenance readjustments shall be provided as part of the technical manual.

3.5.3 Antenna installation site characteristics. The antenna shall be designed such that installation site grading will not be required except for the following reasons:

(a) The overall ground slope exceeds plus or minus 5 degrees from horizontal within a 225-foot radius from the geometric center of the antenna (at ground level), or
(b) The ground roughness exceeds plus or minus 4 feet within the first 100-foot radial distance or exceeds plus or minus 0.04R feet within the next 125-foot radial distance from the ground level geometric center of the antenna, where R is the radial distance.

3.5.3.1 The antenna shall remain within the maximum VSWR requirements specified in MIL-A-28768 when installed over any ground ranging from flat and level to the limits specified in 3.5.3. The pattern requirements specified in 3.6.4 may vary, since the pattern specifications apply to flat, level ground.

3.5.3.2 Installation instructions. The installation instructions shall thoroughly discuss the incremental effects of the ground slope and roughness on the antenna's performance such that installation design personnel may trade off antenna performance against site preparation costs. This information shall be presented in increments of at least 1 degree of levelness and 12 inches of roughness within the limits of 3.5.3.

3.5.3.3 Materials. All necessary materials, such as longer catenaries and guy assemblies, additional anchors and attachment hardware which are necessary to allow installation over any site ranging from flat and level to the limits specified in 3.5.3 shall be supplied.

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3.6 Electrical performance. The antenna shall comply with the electrical design requirements of MIL-A-28768 and the additional performance requirements specified below. Unless otherwise specified, the antenna shall be assumed to be over perfectly conducting earth. Unless otherwise specified, all performance parameters shall be met individually, simultaneously, and in any combination.

3.6.1 Power. The antenna shall be capable of handling the power specified in MIL-A-28768, as classified therein.

3.6.2 Frequency range. The antenna shall be capable of operating without adjustment over the entire frequency range indicated by the type specified (see 1.2).

3.6.3 Gain. The antenna gain at the beam maximum shall not be less than +5 dB (+6 dB if polarization H is specified; see 1.2), above an isotropic radiator in free space, when measured in accordance with the procedures described in 4.3.

3.6.4 Pattern. The following requirements shall be met for frequencies up to 15 MHz. Far field conditions shall be assumed.

3.6.4.1 The beam maximum shall occur at any elevation angle between 40 and 90 degrees. However, gain shall not fall below +2 dB (above isotropic) anywhere between 40 and 90 degrees of elevation for all directions of azimuth; except that the gain may decrease to not less than +1 dB (zero dB for polarization H only) for selected solid angles totaling not more than $\pi/4$(pi/4) steradians.

3.6.4.2 The magnitude of the vertically polarized rf energy available for ground wave radiation at the zero to 1 degree elevation angles, for any given direction of azimuth, shall be at least 15 dB below the beam maximum in that same direction of azimuth, except that it may be as low as 11 dB for frequencies below 1.5 times the low frequency limit for the type specified in 1.2.

3.6.5 As far as practical, the location of the beam maximum at frequencies above 15 MHz shall be in the vicinity of the 30 degree elevation angle.

3.6.6 Polarization. The antenna shall be capable of transmitting or receiving radio waves of the polarization specified by the classification (see 1.2). If circular polarization is specified, the ellipticity in any plane perpendicular to any straight line extending from the antenna to that plane shall be 4 dB plus or minus 3 dB. The polarization requirements stated herein apply only to that portion of the pattern which is defined by 3.6.4.1 (that is, between 40 and 90 degrees of elevation for all azimuths). When circular polarization is specified, the antenna polarization shall be capable of being manually changed from the specified sense of rotation to the opposite sense during installation. Instructions for making the change and the reasons for the change (see 6.1.1) shall be contained in the technical manual.

3.6.7 Efficiency. When installed above imperfectly conducting earth, the efficiency of the antenna as defined in MIL-A-28768 shall not be less than 90 percent (72 percent when the effects of VSWR are considered).

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3.6.8 Diversity. When specified in the procurement document (see 6.2), receiving antennas (power 0 per MIL-A-28768) shall be designed to allow the antenna to be converted for use with a polarization diversity receiving system. This capability shall require only the installation of an additional matching unit to provide the 2 channels needed for diversity reception. The additional matching unit shall not be supplied. Detailed conversion instructions shall be included in the technical manual. The polarization requirements specified above shall not apply after conversion.

3.7 Ground system. The antenna shall not be dependent upon an artificial ground system in order to comply with the requirements of this specification. Hence, such a ground system shall not be supplied.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions shall be in accordance with MIL-A-28768 and as specified herein.

4.2 First article inspection. In addition to the inspection requirements of MIL-A-28768 the following inspections shall be conducted:

(a) Verification of size requirements of 3.3.2/
(b) Verification of wind and ice loading conditions of class required to comply with the requirements of 3.4.1/
(c) Verification of rotational stability requirement of 3.5.1.1/
(d) Verification of compliance to the installation grading requirements of 3.5.3.2/
(e) Verification of compliance to the requirements of 3.6.3 through 3.6.6 in accordance with MIL-A-28768 and the additional procedures of 4.3.1 when a fractional scale model is required.

1/These requirements may be demonstrated by engineering analysis.
2/This requirement may be demonstrated during installability test.

4.3 Inspection procedures. The following inspection procedures supplement those specified in MIL-A-28768.

4.3.1 Gain and ellipticity measurements. Gain measurements shall be made using a linearly polarized dipole. For circularly polarized antennas, these measurements shall be made along both the major and minor axis and the gain then determined by the root-sum-square method. The ratio of the major and minor axis measurements expressed in dB shall be used to determine compliance with the ellipticity requirements.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be as specified in MIL-A-28768.

6. NOTES

6.1 Intended use. The antennas covered by this specification are intended for short range (0 to 500 miles) communication via the ionosphere only, and line of sight air to ground communications. Significant improvement in S/N ratio and

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reliability is obtained over ground wave antennas, except when the entire path is over seawater; then destructive interference makes these antennas unreliable.

6.1.1 Type and polarization. In selection of type and polarization, the following shall apply:

(a) Ionospheric propagation is limited to frequencies in the 2 MHz to 15 MHz range for reliable communications over the ranges specified. Frequencies above 15 MHz are predominately for longer skywave paths and line of sight communications.

(b) The use of circularly polarized antennas at either or both circuit terminals provides a significant protection against Faraday fading. Diversity reception provides even further protection. In selecting the sense of polarization, the following rules shall be applied, otherwise significant degradation will result.

(1) If antennas at both ends of the circuit are circularly polarized, each must be in the opposite sense of the other.
(2) For transmitting applications in the northern hemisphere and receiving applications in the southern hemisphere, right hand circular polarization (RHC, see 1.2) must be used.
(3) For transmitting applications in the southern hemisphere and receiving applications in the northern hemisphere, left hand circular polarization must be used.

(c) Linear polarization may be used (that is, there is no significant advantage to using circular polarization at either end of the circuit) when less omnidirectionality can be tolerated and any one of the following conditions exist:

(1) Circuits are less than 10 degrees of latitude from the earth's equator and the direction of polarization is along magnetic north-south, or
(2) Diversity reception is utilized

6.1.2 Class 2 is assumed in this specification. Class 2 is intended to cover generally encountered service conditions. If unusually light or unusually severe environmental conditions are known to exist, the class must be specified accordingly (see 6.2).

6.2 Ordering data. In addition to the items specified in MIL-A-28768, procurement documents should specify the following:

(a) Type and polarization (see 1.2)
(b) Diversity, if required (see 3.6.8)
(c) Class, if other than class 2 is required (see 3.4)

6.3 Definitions.

Right hand and left hand circular polarization. Circular polarization is right hand when the direction of rotation of the electric field vector is as indicated by the curvature of the fingers of the right hand when the thumb is

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pointed in the direction of propagation; and is left hand when the thumb and fingers of the left hand are used. This definition is consistant with that of the IEEE.

6.4 Other notes. The notes contained in MIL-A-28768 are applicable to this specification.

6.5 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity
Navy - EC

(Project 5985-N395)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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